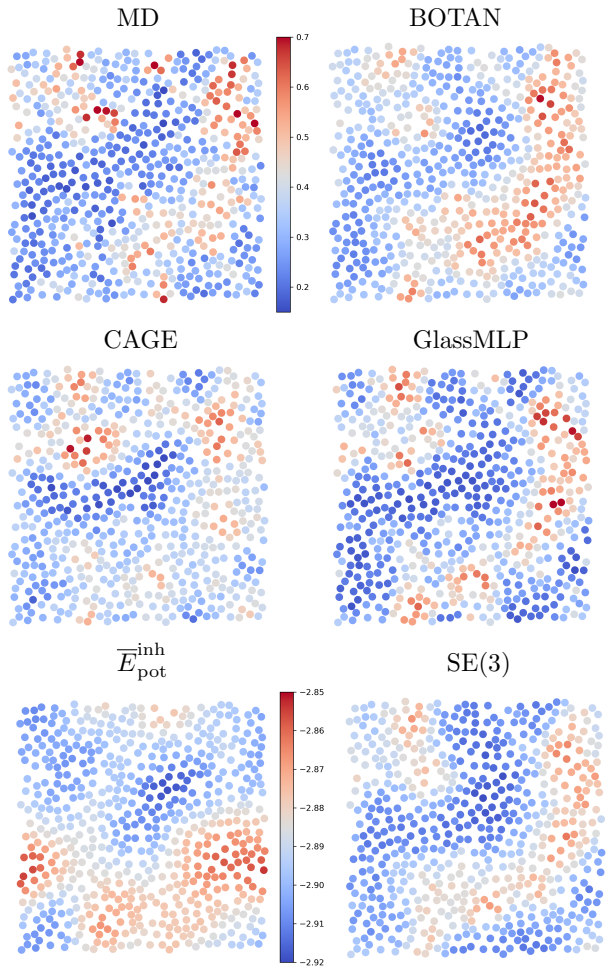

FIG. S4. Snapshots of an exemplary configuration of the KA2D system (shown are only type A particles). Each particle is coloured according to its propensity of displacement $\mathcal{R}$ at $t = \tau_\alpha$, as calculated from MD simulations or predicted from various ML techniques (see Fig. 3 in the main text for the respective Pearson correlations). Each snapshot has the same color bar, except $\overline{E}^{\rm inh}_{\rm pot}$.

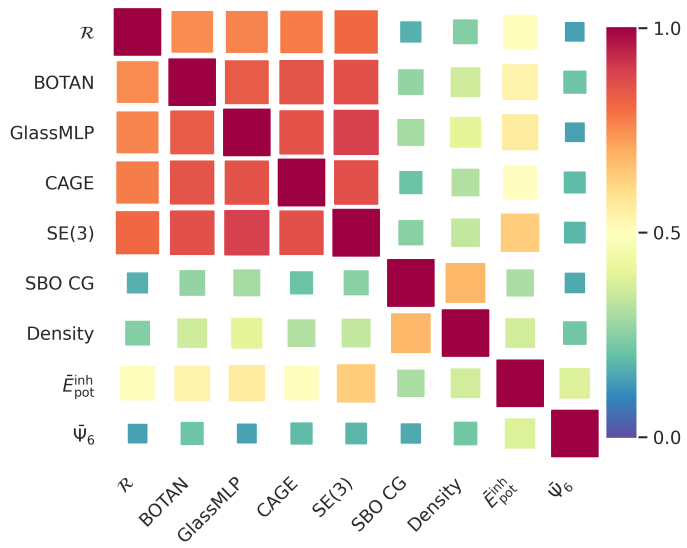

FIG. S5. Pearson cross-correlations $\rho_{\rm P}(X, Y)$ between different ML models and structural descriptors in the KA system (3D) at $t = \tau_\alpha$.

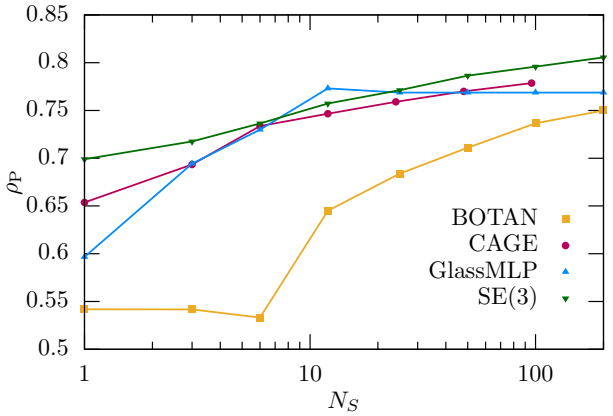

FIG. S6. Performance of network, as quantified by the Pearson correlation coefficient $\rho_{\rm P}$, trained with differently sized training sets in the KA system (3D) at $t = \tau_\alpha$. N_S denotes the number of samples, where each sample consists of $N = 3277$ particles.

E. Cross correlations

To study connections between the different models, we also investigate the cross correlations between the different ML techniques and other structural descriptors. Unsurprisingly, the strongest cross correlations are between the various ML models, since they all correlate strongly with propensity. The strongest correlation can, in fact, be detected between SE(3) and GlassMLP (see Fig. S5). This can be rationalized by the observation that both techniques also have the strongest correlation with the coarse-grained potential energy, $\overline{E}^{\rm inh}_{\rm pot}$. This could be expected since both techniques receive the potential energy as explicit input.

Another strong cross correlation is observable between the unsupervised SBO CG technique and density, showing that these two descriptors are strongly linked. This indicates that the strongest asymmetries in the structure emerge from density fluctuations.

F. Learning curves

An important feature of ML models is also how they cope with limited training data. In standard fitting procedures, one would usually expect that the more fitting parameters a model has, the more training data is required to achieve a certain accuracy. This 'rule of thumb'

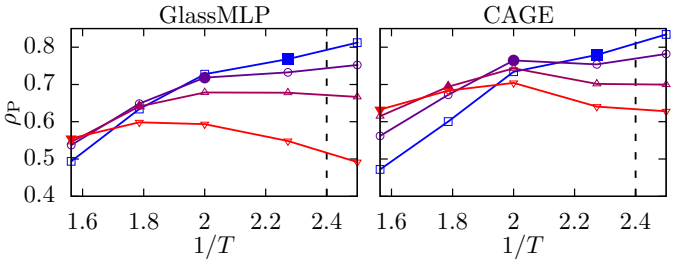

FIG. S7. Transferability in temperature T of trained networks in the KA system (3D) at the structural relaxation time τ_α. The figures is identical to Fig. 6 in the main text, just featuring an additional temperature $T = 0.4$.

has, however, been repeatedly questioned for deep networks, which often perform well even when overparameterized. When investigating Fig. S6 we observe that indeed SE(3) already performs excellent, even with only $N = 3277$ data points, despite having > 50k fitting parameters. Furthermore, the increase in performance is very continuous, implying that the network might perform even better with > 400 independent structures (which was the maximum of the dataset available). The learning curves of CAGE and BOTAN closely resembles SE(3) just with a very small reduction in performance.

Contrarily, the learning curve of GlassMLP starts at a significantly smaller Pearson correlation, but quickly catches up with SE(3) to slightly surpass its performance at around $N_S = 10$ (i.e. 32770 data points). However, the performance of GlassMLP does not further improve for larger $N_S > 20$. It might be necessary to increase the number of structural input descriptors to improve GlassMLP in situations where much training data is available. It should be mentioned that there is an important difference between the training procedures of SE(3) and the other models. While GlassMLP uses for $N_S = 1$ only twice the number of epochs for training (i.e. $N_e = 600$), since no improvement in performance can be observed beyond this point, SE(3) requires a significant increase in epochs for small N_S: $N_e(N_S) = 8 \cdot 10^4/N_S$. This implies that for $N_S = 1$, the training time of GlassMLP is roughly 5 seconds, while for SE(3) it remains roughly 11 hours.

G. Transferability

We have investigated in detail the transferability of trained networks in the benchmarking section V of the main text. Here, we study for two models the performance of GlassMLP and CAGE when applied to an even lower temperature $T = 0.4$, taken from the dataset presented in Ref. [S17]. Different from the main KA dataset we only have $N_S = 12$ individual structures available, but use $N_R = 100$ different isoconfigurational replicas. Due to the larger number of replicas we expect the Pearson correlations to be slightly larger than for $N_R = 32$. The results in Fig. S7 indeed confirm this expectation, nevertheless, they also highlight the strong transferability of the models. A network trained at $T = 0.44$, with a relaxation time an order of magnitude smaller than at $T = 0.4$, CAGE can predict structural relaxation with a Pearson correlation of up to $\rho_P = 0.84$. We believe that this is an important result towards using transferability to study glassy liquids at extremely low temperature.

H. Bond-breaking propensity in the KA system (3D)

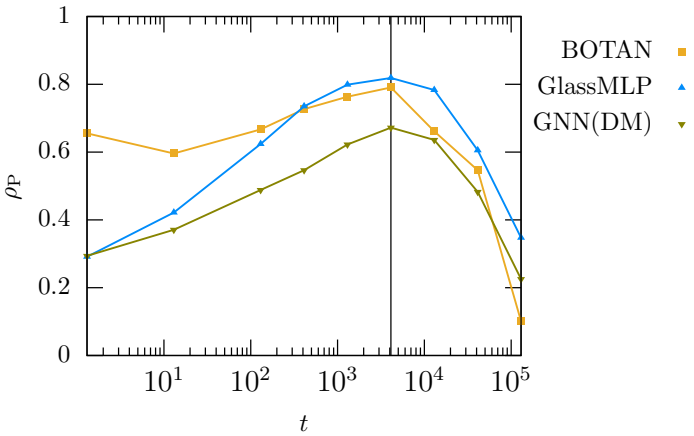

FIG. S8. Pearson correlation ρ_P with the ground truth in the KA system (3D) at $T = 0.44$ for various different techniques. Different from Fig. 3, here the dynamical variable is the bond-breaking propensity. The vertical line marks the structural relaxation time τ_α. GNN(DM) refers to the original GNN [S7].

Finally, we also briefly investigate the performance of ML models and other structural descriptors in predicting the bond-breaking propensity in the KA system (see Fig. S8). While the predictability for small times is much weaker than for the propensity of displacements, $\mathcal{R}$, analysed in Fig. 3, we observe better performance for longer times around the structural relaxation time scale $t = \tau_\alpha$. This observation is identical to the discussion in Sec. V of the main text after Fig. 5 for the bond-breaking propensity in the KA2D system. We further find that both BOTAN and GlassMLP significantly outperform the original GNN proposed by DeepMind ([S7], GNN(DM)). This shows the great advancements in the field within the past three years. We hope that this roadmap further fuels the development and application of ML techniques to analyse glassy liquids.

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